Basic Principles and Operation of Metal Oxide Semiconductor Field Effect Transistor - MOSFET

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Outline

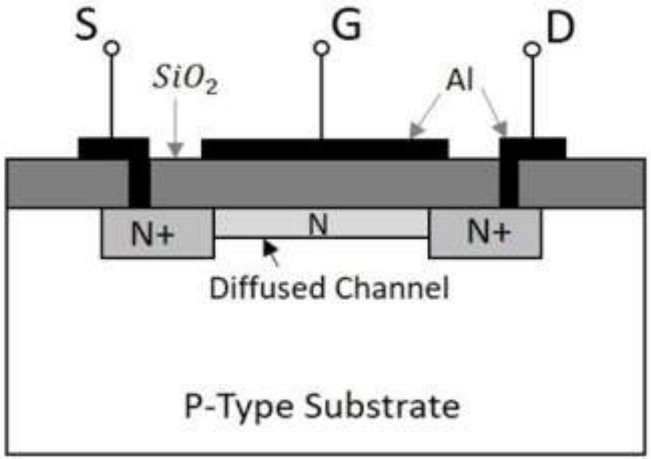
- Introduction to MOSFET.
- Types of MOSFET
- Operation of MOSFET
- Literature review and ongoing research activities with MOSFET technology.
- Applications of MOSFET
- Conclusion

Introduction to MOSFET

A Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is a voltage-controlled semiconductor device used for switching and amplification in electronic circuits.

MOSFET Structure

- **Source (S)**: Terminal through which carrier enters.
- **Drain (D)**: Terminal through which carrier exits.
- **Gate (G)**: Controls the flow of carriers between Source and Drain.

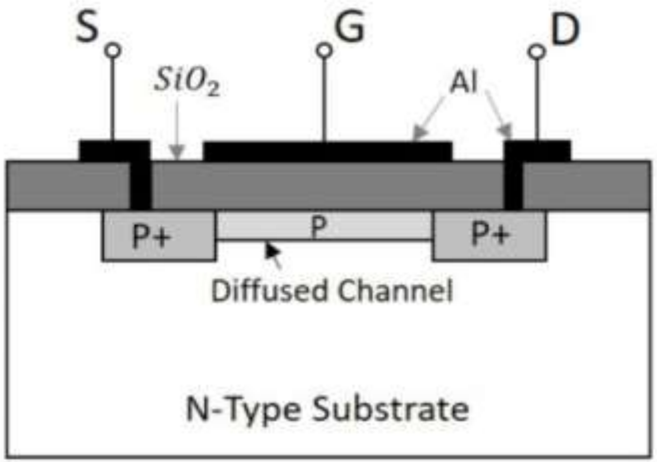


Structure of N-channel MOSFET

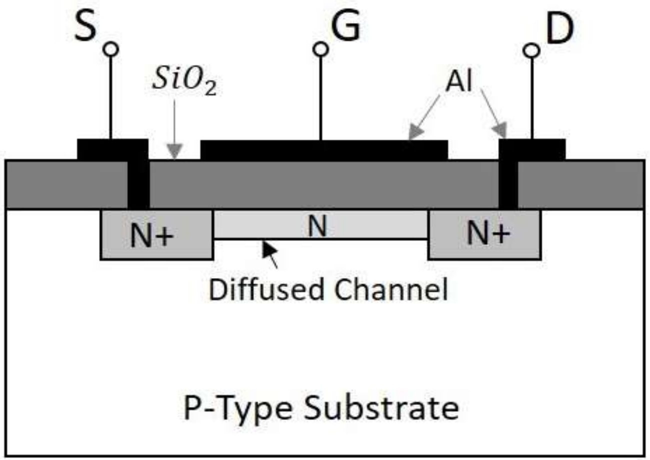
Structure Layers:

- **Gate**: Made of conductive polysilicon.
- **Oxide Layer (SiO_2)**: Insulating material separating the Gate from the semiconductor channel.
- **Channel**: Conductive semiconductor region, either n-type or p-type.

Types of MOSFET



Structure of P-channel MOSFET

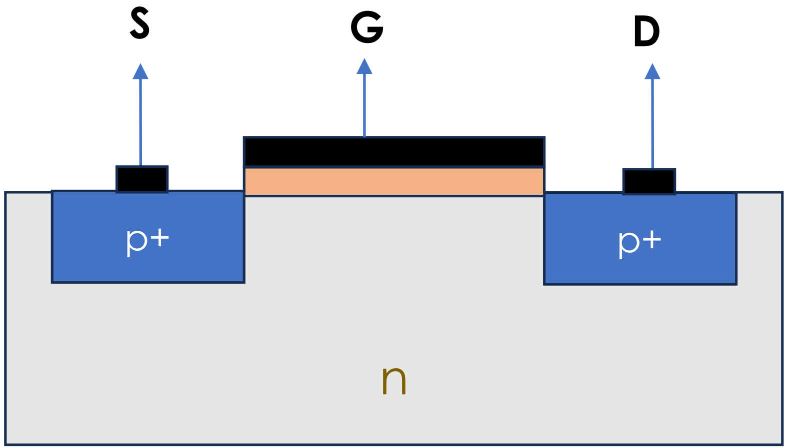


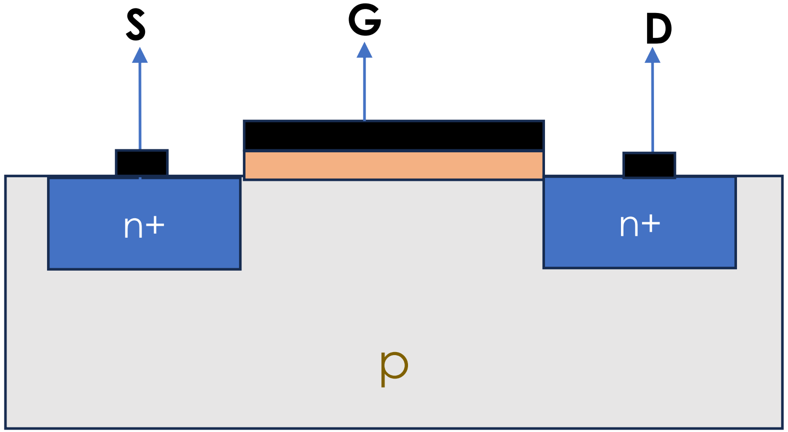
Structure of N-channel MOSFET

Depletion type MOSFET

Types of MOSFET cntd.

Enhancement-type MOSFET





- When $V_{GS} < V_{th}$, holes flow from Source to Drain, allowing current conduction
- **Source**: Positive terminal
- **Drain**: Negative terminal
- **Gate**: Controls current flow between Drain and Source.

- When $V_{GS} > V_{th}$ electrons flow from Source to Drain, allowing current conduction
- **Source**: Negative terminal
- **Drain**: Positive terminal
- **Gate**: Controls current flow in opposite direction.

Operation of e-nMOS

Threshold Voltage (V_{th}): Minimum Gate voltage required to turn the MOSFET on.

Regions of Operation:

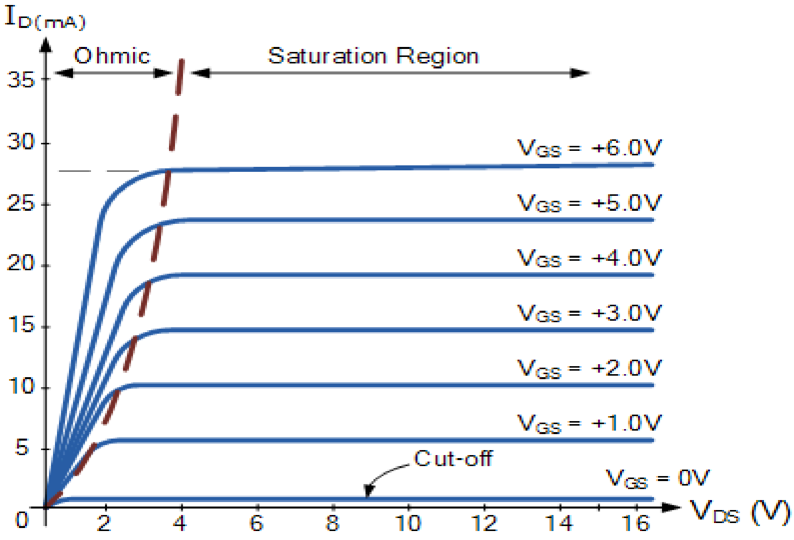
- **Cutoff Region**: No current flow,
  $I_D = 0 \quad \text{for} \;\; (V_{GS} < V_{th})$.

- **Linear Region**: Current flows, MOSFET behaves like a variable resistor ($V_{GS} > V_{th}$ and $V_{DS} < V_{GS} - V_{th}$).

$$I_D = k_n(2(V_{GS} - V_{th})V_{DS} - {V_{DS}}^2)$$

- **Saturation Region**: MOSFET is fully on, controlled by Gate voltage ($V_{GS} > V_{th}$ and $V_{DS} > V_{GS} - V_{th}$).

$$I_D = k_n(V_{GS} - V_{th})^2, \qquad k_n = \frac{\mu_n \varepsilon_{ox} W}{2T_{ox}L}$$



Key performance parameters of MOSFET

Parameter	Description	Impact on Performance
Threshold voltage $(\boldsymbol{V_{th}})$	Minimum Gate voltage for conduction	Affects switching behavior and leakage current.
Drain Current $(\boldsymbol{I_D})$	Current between Drain and Source	Determines how much current the MOSFET can handle
Transconductance $(\boldsymbol{g_m})$	Rate of change of $\boldsymbol{I_D}$ w.r.t. $\boldsymbol{V_{GS}}$	Higher $\boldsymbol{g_m}$ improves amplification and switching speed.
Gate Capacitance $(\boldsymbol{C_{GS}, C_{GD}})$	Capacitance between gate-source (Cgs) and gate-drain (Cgd)	Lower capacitance improves switching speed.
Subthreshold Slope **(SS)**	Slope of current-voltage characteristic in the subthreshold region.	A lower slope enhances low-power operation.
Power Dissipation (**Pd**)	Power lost as heat due to resistive and switching losses.	Minimizing power dissipation is crucial for energy efficiency.
Cut-off frequency $(\boldsymbol{f_t})$	Measure of the frequency at which the MOSFET's **current gain** falls to 1	At $\boldsymbol{f_t}$, the MOSFET's gain (or transconductance) starts to drop significantly.

Approaches toward performance development

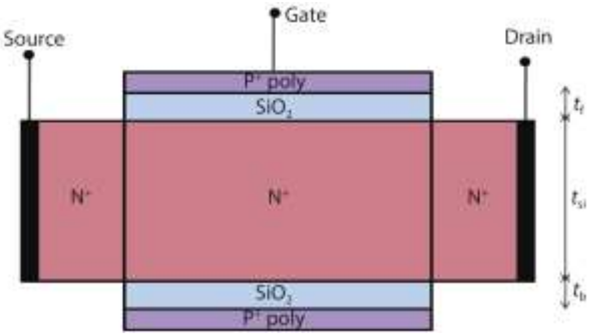
Since the transistor technology node is downscaling, IC chip contains increased number of transistor in it. However, this increased density also amplifies undesirable phenomena, such as short-channel effects (SCEs) which weaken the gate's control over the channel.

Some of the techniques that are adopted to face these challenges:

- Optimized and novel device structure like double gate, triple gate, FinFET, GAA-FET.
- Junction less MOSFET
- High mobility channel material
- High-k gate dielectric
- Efficient fabrication technology

Approaches toward performance development cntd.

(a)



Tripathi et. al. 2020

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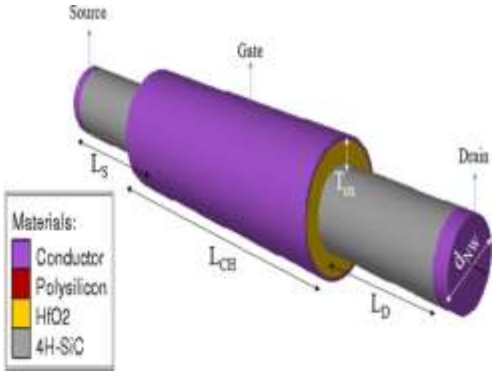

Orouji et. al. 2011

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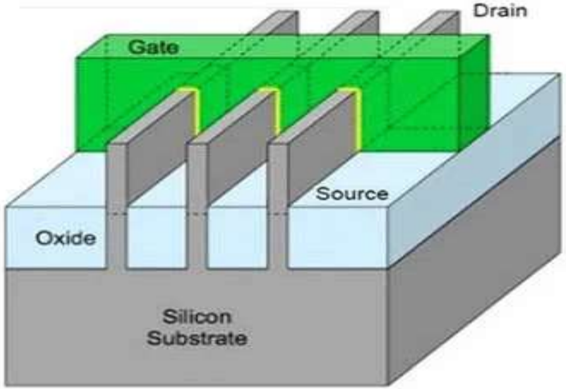

Image courtesy of Intel 2010

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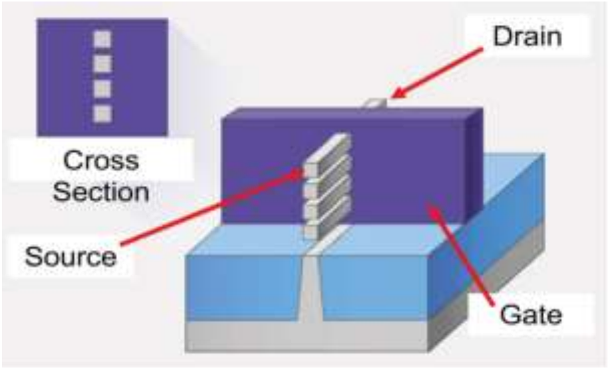

Lin j. et.al. 2022

Applications

- ✓ **Digital Circuits**: CMOS logic gates, microprocessors.
- ✓ **Amplifiers**: Used in analog signal amplification
- ✓ **High-Speed Digital Circuits:** Microprocessors, FPGAs, high-speed memory interfaces.
- ✓ **Switching**: Power control in power supplies, motor control circuits.
- ✓ **Sensing and Detection:** Label-free biomolecule detection, gas sensors.

Summary

- MOSFETs are essential components in digital, analog, and power electronic systems.
- Operation is controlled by the Gate voltage, allowing for both switching and amplification.
- MOSFETs are widely used in CMOS technology, which forms the foundation of modern integrated circuits.

Thank You!